

User Feedback System for Canvas

Adwitia Aoishee, Clarissa Kuntzman, Alondra Rosales, Nooreen Khan

Introduction

The primary objective of this project was to design and implement a user feedback system tailored for Canvas. The goal was to allow users (students, faculty, and staff) to easily report issues, suggest improvements, and rate their experience using Canvas. To achieve this, we developed a web-based interface that prioritized usability, accessibility, and simplicity.

Project Timeline and Design Process

The development process spanned nine weeks and followed a structured timeline. We started with research and planning, then moved on to design and prototyping. After that, we built the site, tested it, and made improvements based on user feedback. Throughout the process, we made sure to reference the Four Pillars of HCI to ensure that our design aligned with HCI concepts.

HCI Framework

We applied the 4 Pillars of HCI to guide development:

1. User-Interface Requirements - Ensuring features matched user needs.
2. Guidelines, Documents & Process - Following best practices in interface design.
3. User-Interface Software Tools - Utilizing technologies for development and testing.
4. Expert Reviews & Usability Testing - Conducting structured evaluations.

Key Design Principles

Accessibility

To ensure inclusivity, we included a variety of features to tailor to different users' needs. We created a Frequently Asked Questions (FAQs) section to provide users with answers to common concerns before they begin filling out the form. We also added dark mode and high contrast toggles. While these can be used for visual preference, they're especially important for users with visual impairments. For example, dark mode reduces eye strain and is great for low-light settings. Further, high contrast mode helps users with low vision or contrast sensitivity by making text stand out more clearly against the background. On top of that, we included a font size adjuster that allows users to pick a text size that's most comfortable for them. This is especially helpful for people with limited vision or reading challenges who might otherwise need to zoom their entire browser. Throughout the portal, we use clear labels and mark required fields with a red asterisk. This makes it easier to follow along and is more screen reader-friendly too. Finally, we chose simple fonts and high-contrast color combinations that make everything easy to read in any mode. All of these design choices fall under the HCI pillar of User-Interface Requirements, since we're prioritizing real user needs.

Logical Layout

We designed the layout to prioritize ease of navigation. At the top of the page, we included a compact navigation bar. This gives users quick access to helpful tools like the FAQs page and accessibility settings. Whether someone needs to toggle between light and dark mode, increase contrast, or adjust the font size, they can easily customize the site to their preferences. As users move into the survey, questions are presented one at a time. This reduces cognitive load, which helps prevent users from feeling overwhelmed. Instead of scanning through a wall of questions, they get a more guided survey and take it one question at a time. We also added a progress indicator at the top of the form. It shows which question they're currently on and how many are left, which helps users pace themselves and feel a sense of progress. This allows users to get an idea of how long the survey will take them. Additionally, we included "Previous" and "Next" buttons to let users easily go back if they want to review or change an answer, which adds to the form's flexibility and usability. The entire layout is designed to be mobile responsive. Whether someone is using a desktop, a tablet, or a phone, the form adjusts to fit their screen, so it's easy to use on any device. This was key for ensuring the form is inclusive and accessible no matter how someone chooses to access it. All of these choices reflect the HCI pillar of

User-Interface Requirements because they aim to reduce friction and make it easy for users to complete their tasks with minimal confusion.

Question Design

Another important part of our design was the way we structured the survey questions. From the start, we focused on using clear, direct language that any user can easily understand. This helps reduce confusion and makes it easier for people to engage. Most of the questions are formatted as multiple choice, dropdowns, or sliders. This means users don't need to type out long responses, unless they want to. We limited how much manual typing was required to keep the process quick and approachable while reducing errors since typed answers are more error prone. For example, we included a dropdown menu asking about technical issues, a multi-select list for commonly used features, and a slider for rating how easy navigation is. Each of these are easy to answer immediately and encourages people to keep going. We also made sure that required questions are clearly marked with a red asterisk so users know exactly what they need to fill out and what they can skip. The goal behind this question design was to make the survey low-effort, low-stress, and easy to complete without overwhelming the user. This supports the HCI principle of User-Interface Requirements, to make interactions efficient and user-friendly.

Submission Flow

The submission experience was made clear and user-friendly. This key part of our design focused on what happens after a user completes the form. First, we built in error prevention to make sure users don't accidentally skip over required questions. If they miss something, the field is outlined in red, and they won't be able to move on until the issue is resolved. This visual cue keeps users on track without breaking their focus. After the form is successfully submitted, users are immediately taken to a confirmation screen. It thanks them for their input and lets them choose to either submit another response or return to the main Canvas page. This kind of feedback is important for building trust in the system, so users know their response was received. Another helpful feature we included is a 10-minute response editing window. After submitting, users have the ability to go back and change their answers if they realize they made a mistake. This adds flexibility and supports user control, which is a core HCI principle. Overall, the submission process is designed to feel smooth, informative, and complete. It connects back to the HCI pillar of User-Interface Requirements, by reducing frustration and supporting task completion. It also

relies on the User-Interface Software Tools pillar to make those alerts, confirmations, and redirects work together.

Tools and Technologies

The system was built using:

- Frontend: HTML, CSS, JavaScript
- Backend: Basic server-side logic
- Hosting: Deployed on w3spaces/w3schools
- Data Collection: Survey responses and analytics collected via the platform.

Usability Testing

Usability testing was conducted over two weeks and involved 10 users (2 groups of 1 teacher + 4 students.) We collected data using think-aloud protocols and post-survey interviews. Participants provided positive feedback about our “one question at a time” approach to the survey, stating how they found it easy to follow along. They also appreciated the accessibility features, especially dark mode and font size options. After receiving feedback, we introduced a feature allowing users to edit submissions for 10 minutes.

Future Goals and Improvements

For our future goals and improvements, we would like to add multilingual support to expand accessibility across languages, a follow-up mechanism to notify users when their feedback is addressed, and a response summary & admin dashboard to summarize top feedback trends for Canvas support staff.

Conclusion

The project successfully reached our goal of delivering a functional, user-friendly feedback system for Canvas. Every design decision we made was driven by the principles of HCI. We prioritized simple question design, a clean and mobile-friendly layout, accessibility tools, and a smooth submission flow that guides users from start to finish. Through iterative

design, accessibility consideration, and user testing, we created a tool that can meaningfully improve user experience on educational platforms like Canvas.